



Ts. Dr. Aliza binti Che Amran

Control Systems • Robotics • Intelligent & Sustainable Engineering Systems



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A. PROFESSIONAL PROFILE SUMMARY

Senior Academic and Institutional Leader with over 20 years of experience in electrical engineering, specializing in **Control Systems, Robotics, and Intelligent Systems** with strong real-world impact. Demonstrated success in institutional transformation, including a 100% accreditation success rate and contributions to Universiti Teknikal Malaysia Melaka (UTeM) inclusion in the QS World University Rankings 2024.

Led the development of national TVET frameworks (NOSS P853-001-5:2017) and pioneered competency-based postgraduate education aligned with the Malaysian Qualifications Framework. Recognized for educational innovation through SULAM (*SULAM-Service Learning Malaysia, University for Society*) initiatives and for delivering impact-driven projects, including a RM100,000 UniMADANI-funded hybrid composting system with recognition from SWCorp Malaysia. Combines engineering, policy, and community engagement to deliver sustainable, human-centered solutions.

B. AREA OF EXPERTISE

- **Advanced Engineering & Research Systems**
 - a. Control Systems Design, Robotics
 - b. Human-Centered & Autonomous Engineering Systems
 - c. Sustainable Waste Management & Circular Economy Systems
- **Academic Leadership, Governance & Quality Systems**
 - a. Programme Accreditation & Compliance (MQA, ISO 9001:2015)
 - b. Outcome-Based Education (OBE) & Curriculum Transformation
 - c. Institutional Quality Assurance & Audit Systems
 - d. Academic Community Leadership & Stakeholder Engagement
- **Pedagogical Innovation & Academic Development**
 - a. Competency-Based Education (TVET, NOSS Framework Development)
 - b. Service-Learning (SULAM) & Experiential Learning Design
 - c. Academic Staff Development, Coaching & Mentoring
- **Research Translation, Impact & Stakeholder Integration**
 - a. Multi-Stakeholder Project Leadership (Government–Academia–Community)
 - b. Grant Acquisition & Project Management
 - c. Policy–Technology Integration for Societal Impact
- **Strategic, Analytical & Professional Capabilities**
 - a. Data-Driven Decision Making & Evidence-Based Evaluation
 - b. Root Cause Analysis & Continuous Improvement (QMS Systems)
 - c. Technical Writing, Research Publication & Knowledge Dissemination

C. AWARD AND RECOGNITION

1. **Excellent Service Award Recipient** (UTeM)
 - Based on performance evaluation of the year 2008 – conferred on 25th May 2009
 - Based on performance evaluation of the year 2017 – conferred on 8th March 2018
2. **Finalist**, UTeM Special Group Award at the Academic Awards Ceremony 2019, *TVET Research and Innovation Group (TRIG) (Team Leader)*
3. **Finalist**, UTeM Special Group Award at the Academic Awards Ceremony 2019, *STEM@UTeM (Team Member)*
4. **Finalist**, Best e-SULAM Concept Project at the e-SULAM Innovate 2021 Final Competition (Concept Category), *“After-School e-Tutoring”*
5. **Bronze**, UTeMEX 2017, *“Technical Lecturer Professional Development Programme”*
6. **Bronze**, IDEX 2019, *“Development of Biomass Stove Using Thermoelectric for Commercial Use”*
7. **Silver**, IDEX 2020, *“Smart Power Consumption Monitoring System”*
8. **Bronze**, IDEX 2020, *“Development of Foot Sole Pressure Pads for Rehabilitation Purposes”*
9. **Gold**, UTeM IDP Day 2021
10. **Best Paper Award (Track: STEM Education)**, Innovative Teaching & Learning Research Day 2021, *“Online STEM Mentoring Program for Service-Learning Experience Among First Year Engineering Technology Undergraduates”*
11. **Silver Poster Award**, Innovative Teaching & Learning Research Day 2021
12. **Silver**, iIDEX 2025, *“Development of an Electric Food Waste Composter Machine for Household”*

D. ACADEMIC QUALIFICATION

- 2013 | Doctor of Engineering (Physics, Electrical and Computer Engineering)
 - Yokohama National University, Yokohama | *Research field: Bipedal robot walking trajectory*
- 2006 | Master of Electrical and Computer Systems Engineering (Electrical Engineering)

- Monash University, Melbourne | *Master by Coursework*
- 2003 | Bachelor of Electrical and Electronics Engineering (Control and Instrumentation)
 - Universiti Teknologi Petronas, Perak | *FYP topic on Automatic Load Shedding System*

E. ACADEMIC APPOINTMENT

Henan Polytechnic, Zhengzhou, Henan, China

- April – May 2026 | Visiting Lecturer | International Cooperation and Exchange Department

Universiti Teknikal Malaysia Melaka (UTeM)

- 2015 – 2025 | Senior Lecturer | Fakulti Teknologi dan Kejuruteraan Elektrik
- 2013 – 2015 | Senior Lecturer | Fakulti Kejuruteraan Elektrik
- 2007 – 2013 | Lecturer | Fakulti Kejuruteraan Elektrik

Kolej Universiti Teknikal Malaysia Melaka (KUTKM)

- 2005 – 2007 | Lecturer | Fakulti Kejuruteraan Elektrik
- 2004 – 2005 | Tutor | Fakulti Kejuruteraan Elektrik

F. PROFESSIONAL MEMBERSHIP AND CERTIFICATION

- Board of Engineers Malaysia (BEM) | Graduate Engineer (G148847 A) (Electrical branch) since April 2006
- Malaysia Board of Technologists (MBOT) | Graduate Technologist (GT18020083) in Electrical & Electronics (EE) Technology | Professional Technologist (PT18030464) in EE Technology
- ISO 9001:2015 QMS Lead Auditor certified by Exemplar Global

G. LEADERSHIP APPOINTMENT (*selected*)

Deputy Director, Centre of Teaching and Learning (UTeM) | 2013 to 2015

- Led academic enhancement initiatives in curriculum design and pedagogical innovation aligned with best practices that involves engineering, engineering technology and non-engineering education (OBE)
- Drove integration of educational technologies and hybrid learning strategies to improve teaching delivery and student engagement (e.g. Massive Open Online Course (MOOC), Blended Learning, etc.)
- Spearheaded structured staff development programmes including training, coaching, and new faculty induction
- Strengthened teaching quality assurance through systematic evaluation, feedback mechanisms, and performance improvement interventions

Project Leader, Development of NOSS-Based TVET Framework for Technical University Lecturers (P853-001-5:2017)

- Secured and led a top-down university grant RM84,000 (Revitalizing Teaching and Learning, 2015), initiating the development of the Technical Academician Competency Profile (TACP) as a foundation for competency-based academic practice.
- Subsequently advanced the work under a NOSS development grant by the Department of Skills Development (DSD), translating TACP into a formalized NOSS-based TVET framework (P853-001-5:2017, TVET Implementation and Development) aligned with the Malaysian Qualifications Framework (MQF).
- Contributed to the advancement of national TVET qualification pathways, supporting the elevation of TVET Section recognition within MQF 2.0 up to Level 8 (approved by MQA Council, December 2017).
- Translated the NOSS framework into a postgraduate programme, the Master of Technical Education and Training, positioned at MQF Level 7 (Academic Section), designed for TVET educators across multiple ministries.
- Pioneered a novel postgraduate curriculum model integrating NOSS competency standards into Master's level education (40 credits), establishing dual academic–professional qualification pathways.
- Enabled graduates to obtain both a Master's degree and eligibility for the Malaysian Advanced Skills Diploma (DLKM) in TVET Implementation and Development (P853-001-5:2017).
- Introduced one of the first NOSS-driven curriculum implementations at postgraduate level in Malaysia, bridging academic and occupational standards.

TVET Document Forensic Taskforce (Ministry of Human Resources (MOHR)) | 2021

- Appointed by MOHR Malaysia to support national TVET quality assurance initiatives
- Analyzed and validated alignment between NOSS P853-001-5:2017 (DSD) and MQA TVET Teaching Competency Guidelines (2021)
- Contributed to strengthening standardization and consistency in national TVET competency frameworks

Coordinator, MESTECC Mentor-Mentee STEM (UM STEM Center) | 2018 - 2020

MySTEM Ambassador Coordinator (Ministry of Higher Education (MOHE)) | 2020

- Led the implementation of STEM mentoring and outreach programmes across selected schools in Melaka, in collaboration with University of Malaya STEM Centre and the Melaka State Edu. Dept.
- Pioneered the integration of STEM outreach programmes with the national Service-Learning initiative (SULAM-Service Learning Malaysia, University for Society), aligning educational activities with community-based learning objectives
- Designed and developed an innovative SULAM programme (After-School e-Tutoring), linking STEM education with real community needs and experiential learning
- Achieved national recognition as Finalist, Best e-SULAM Concept Project, at the e-SULAM Innovate

2021 Final Competition (Concept Category)

- Recognized as a reference point within UTeM for SULAM programme design and implementation, supporting institutional adoption of SULAM as a key KPI under MOHE
- Led the design and implementation of SULAM-based programmes across faculties, strengthening university–community engagement and experiential learning practices

QMS MS ISO 9001:2015 Internal Auditor (UTeM) | 2014-2025

- Conducted annual internal audits to ensure institutional compliance with ISO 9001:2015 standards
- Identified Non-Conformities (NC) and Opportunities for Improvement (OFI), leading root-cause analysis and corrective action verification
- Strengthened process governance, audit integrity, and continuous quality improvement practices across the university
- Developed expertise in institutional quality systems, compliance monitoring, and analytical oversight
- Responsibility: Appointed as an internal auditor to carry out internal audit process as required by the standard. I conducted annual quality audits by reviewing SOPs and performing on-site evaluations to ensure institutional compliance with standards. I was responsible for identifying Non-Conformities (NC) and Opportunities for Improvement (OFI), facilitating root-cause analyses, and verifying corrective actions. Become an expert in process governance and analytical oversight, consistently driving continuous improvement and maintaining the integrity of the university's Quality Management System (QMS).

Deputy Director, Centre for Strategic, Quality & Risk Management (UTeM) | 2021-2023

- Led institutional academic quality governance, overseeing programme accreditation and quality management systems aligned with Malaysian Qualifications Agency (MQA) and ISO MS 9001:2015
- Achieved a 100% accreditation success rate for full and renewal programme evaluations, strengthening institutional compliance and audit readiness
- Contributed to the strategic redesign of institutional performance frameworks supporting global ranking positioning, leading to UTeM's first inclusion in the QS World University Rankings 2024 (1201–1400 band)
- Spearheaded the implementation and enhancement of ISO MS 9001:2015 across the university, strengthening quality assurance processes and governance frameworks
- Led the integration of the Registrar's Office ISO MS 9001:2015 certification into the university-wide Quality Management System (QMS), enhancing system coherence while optimizing administrative costs through consolidation of external audit processes
- Re-engineered feedback analytics mechanisms for staff satisfaction surveys, enabling more accurate, evidence-based identification of performance gaps across departments
- Streamlined the university's quality objectives framework by rationalizing over 100 indicators into 10 high-impact, measurable objectives, significantly improving strategic focus and decision-making effectiveness
- Introduced an objective and evidence-based evaluation system for institutional quality performance, with findings formally presented to and endorsed by the University Quality Council
- Strengthened audit governance by redefining roles and responsibilities of lead auditors, including structured reporting mechanisms to the University Quality Council
- Enhanced cross-faculty coordination in KPI monitoring and risk-based quality improvement initiatives, improving overall institutional performance and governance alignment

Committee Member, Establishment of UTeM APEL Centre | 2022 - 2024

- Contributed to the establishment of the university's Accreditation of Prior Experiential Learning for Credit Award (APEL.C) Center
- Played a key role in the pre-accreditation application process, resulting in official approval by Malaysian Qualifications Agency (valid 7 January 2025 – 6 January 2030)
- Supported the development of assessment frameworks aligned with national APEL standards and practice-based evaluation approaches
- Strengthened institutional capacity in recognizing prior experiential learning and promoting lifelong learning pathways
- Participated in specialized training on APEL.C implementation and practice-based assessment to support center readiness

Project Leader, Electrical Composting System (Komuniti@UniMADANI Grant) | 2024–2025

- Secured and led a RM100,000 Komuniti@UniMADANI grant, awarded by Ministry of Finance (MOF), delivering a community-based electrical composting system within a 4-month implementation period
- Designed an innovative hybrid electrical composting system to optimize energy usage, reduce operational cost, and improve composting efficiency
- Pioneered a phased community adoption model, integrating manual composting practices prior to semi-automated system deployment to enhance user understanding, ownership, and sustainability
- Collaborated with SWCorp Melaka to train and empower local communities in low-cost composting methods using locally available materials, strengthening grassroots capability
- Successfully transitioned the community from manual composting to semi-automated processes, improving efficiency while maintaining process awareness and long-term adoption
- Achieved formal recognition from SWCorp Malaysia and received national media coverage (Selamat Pagi Malaysia and RTM Special Report), demonstrating impact in sustainable waste management and circular economy initiatives

H. RESEARCH GRANTS / PROJECTS – selected

FLAGSHIP PROJECTS (EDUCATION, TVET & IMPACT)

- **Revitalising Teaching and Learning: Development of Technical Academician Competency Profile (TACP)**
Project Leader | 2015 | PJP Top-Down Grant (UTeM) | RM84,000 | Completed
Key Impact: Established a foundational competency framework for technical academicians, subsequently translated into the national NOSS (P853-001-5:2017) and postgraduate programme (SPLT)
- **W2W-4: Circular Economy Implementation through Rumah Kompos Madani @ Pantai Siring**
Main Researcher | 2024 | Komuniti@UniMADANI (MOF) | RM50,000 | Completed
Key Impact: Developed a hybrid electrical composting system with a phased community adoption model; recognised by SWCorp Malaysia and featured in national media (RTM, Selamat Pagi Malaysia)
- **W2W-5: Circular Economy Implementation through Rumah Kompos Madani @ Taman Dato' Abdul Aziz**
Main Researcher | 2024 | Komuniti@UniMADANI (MOF) | RM50,000 | Completed

OTHER RESEARCH GRANTS - selected

- **Green Economy Transformation in Taman Merdeka through IoT-Based Smart Aquaponics**
Co-Researcher | 2025 | Komuniti@UniMADANI (MOF) | RM50,000 | Granted
- **Empowering Circular Economy through Rumah Kompos Madani: Supporting Prison Community Industry**
Main Researcher | 2025 | Komuniti@UniMADANI (MOF) | RM75,000 | Granted
- **ML-1: Community Rehabilitation Programme (PDK) for Stroke Patients through Assistive Technology**
Co-Researcher | 2024 | Komuniti@UniMADANI (MOF) | RM100,000 | Completed
- **Development of Robotic Control System for Composite Trimming of A320 Top Cover Components**
Co-Researcher | 2021 | Industry Grant | RM120,000 | Completed
- **Formulation of Adaptive Algorithm for Stability of Active Suspension System under Chaotic Phenomena**
Co-Researcher | 2019 | FRGS, Ministry of Higher Education Malaysia | RM43,600 | Completed
- **STEM Mentor–Mentee Programme (IEEE UM Student Branch Initiative)**
Coordinator & Team Leader | 2018 | MESTECC | RM63,000 | Completed

I. PATENTS REGISTERED

1. Home Monitoring Mobile Robot with Camera (Copyright) – UTeM Co-Inventor (2016)
2. Mobile Robot Prototype for Interoffice Documents Delivery (Copyright) – UTeM Co-Inventor (2016)
3. Robotic Hand System Via Wire Mechanism (Copyright) – UTeM Co-Inventor (2016)
4. Dexterous Robotic Hand System (Copyright) – UTeM Co-Inventor (2016)

J. SCOPUS-INDEXED PUBLICATIONS – selected

Scopus h-index: 5, Number of Documents: 29, Total citations: 68

1. Amran, Aliza Che, Khairul Haziq Syahmi Khairuhisham, Mohd Farriz Basar, Muhammad Suhaizan Sulong, Addie Irawan, and Ahmad Hadinata Fauzi. "A Smart IoT Framework for Continuous Monitoring and Proactive Prevention of LPG Gas Leaks in Domestic Environments." In 2025 IEEE 16th Control and System Graduate Research Colloquium (ICSGRC), pp. 184-188. IEEE, 2025.
2. Basar, Mohd Farriz, Nur Irrina Balqis Mohd Farriz, Norhaslinda Hasim, Muhamad Faizal Yaakub, Mohd Badril Nor Shah, and Aliza Che Amran. "A Boolean Logic-Based Circuit for Counterfeit Currency Detection: Preliminary Design and Optimization." In 2025 IEEE 16th Control and System Graduate Research Colloquium (ICSGRC), pp. 127-131. IEEE, 2025.
3. Rozali, Sahazati Md, Rozilawati Mohd Nor, Norfariza Ab Wahab, Aliza Che Amran, Saleha Mohamad Saleh, and Muhammad Nizam Kamarudin. "Prototype of electrical generator development based on water flow pressure." TELKOMNIKA (Telecommunication Computing Electronics and Control) 20, no. 2 (2022): 455-461.
4. Sahazati, M. R., N. K. Muhammad, M. S. Saleha, and C. A. Aliza. "Linear quadratic regulator design for vehicle suspension system." ARPN J. Eng. Appl. Sci 17, no. 1 (2022): 67-72.
5. Amran, Aliza Che, Gede Pramudya Ananta, Mohd Ariff bin Mat Hanafiah, Adlan Ali, and Che Ku Nuraini Che Ku Mohd. "Development of the Malaysian skills certification for lecturers in tertiary TVET institutions." Journal of Technical Education and Training 12, no. 1 (2020).
6. Aliza C.A., Ariffin A.Z., Hanipah H., Adlan A., Ariff M.H.M., Gede A.P., "Development of the Malaysian skills certification for TVET lecturers", in International Journal of Advanced Science and Technology, 28 (1), pp. 464 – 477 (2019).
7. Pramudya, G., M. Sedek, F. Shahbodin, A. C. Amran, and S. N. A. Ruslan. "Computer-assisted intervention (CAI) to enhance mathematical learning of autistic students: A case study in Melaka." Humanities & Social Sciences Reviews 7, no. 2 (2019): 196-203.
8. Pramudya, G., M. Sedek, F. Shahbodin, A. C. Amran, and S. N. A. Ruslan. "Computer-assisted intervention (CAI) to enhance mathematical learning of autistic students: A case study in Melaka." Humanities & Social Sciences Reviews 7, no. 2 (2019): 196-203.
9. Ghazaly, Mariam Md, Mohamad Adzeem Mohamad Yuden, and Aliza Che Amran. "Point-to-Point Positioning Control Performance of a 2-DOF Robotic Finger." International Journal of Mechanical &

10. Ghazaly, Mariam Md, Mohamad Adzeem Mohamad Yuden, and Aliza Che Amran. "Motion and Grasping Control Method of 2-DOF Robotic Finger" in International Journal of Mechanical and Mechatronics Engineering, 19 (5), pp. 27 – 32 (2019).
11. Pramudya, Gede, Aliza Che Amran, Muhammad Suyanto, Siti Nur Azrreen Ruslan, Helmi Adly Mohd Noor, and Zuraida Abal Abas. "Assessing Assistive Learning Technologies with Experimental Design." International Journal of Advanced Computer Science and Applications 10, no. 7 (2019).
12. Amran, Aliza Che, Ahmad Muzaffar Abdul Kadir, Ahmad Shukri Zainal, Rozilawati Mohd Nor, Sahazati Md Rozali, Saleha Mohamad Saleh, and Ahmad Hadinata Fauzi. "Analysis of light bulb temperature control for egg incubator design." International Journal of Integrated Engineering 11, no. 4 (2019).
13. Tee, Siau Ping, Mariam Md Ghazaly, Aliza Che Amran, and Irma Wani Jamaludin. "Experimental investigation of a passive quarter car suspension system." ARPN Journal of Engineering and Applied Sciences, ISSN 6608 (1819) (2018).
14. Yuden, M. A. M., M. M. Ghazaly, I. W. Jamaludin, A. C. Amran, and Z. Abdullah. "Optimization design and analysis of an underactuated dexterous robotic hand system." In 2016 International Conference on Robotics, Automation and Sciences (ICORAS), pp. 1-6. IEEE, 2016.
15. Amran, Aliza Che, Naoki Motoi, and Atsuo Kawamura. "Generation of minimal energy joint trajectories during single support phase for 3-dimensional bipedal walking." IEEE Journal of Industry Applications 2, no. 3 (2013): 150-160.
16. Yuden, Mohamad Adzeem Mohamad, Mariam Md Ghazaly, Aliza Che Amran, Irma Wani Jamaludin, Khoo Hui Yee, Mohd Rusdy Yaacob, Zulkeflee Abdullah, and Yeo Chin Kiat. "Positioning control performances of a robotic hand system." Jurnal Teknologi (Sciences & Engineering) 79 (2017): 1-23.
17. Ghazaly M.M., Yew S.K., Abdullah Z., Mohd Ali M.A., Horng C.S., Amran A.C., "Trajectory tracking of a mobile robot system", in Journal of Mechanical Engineering, (Special Issue 1), pp. 207 - 226, (2017).
18. Yudena, Mohamad Adzeem Mohamad, Mariam Md Ghazaly, Aliza Che Amran, Irma Wani Jamaludina, Khoo Hui Yee, Mohd Rusdy Yaacob, Zulkeflee Abdullah, and Yeo Chin Kiata. "Positioning control performances of a robotic hand system." Jurnal Teknologi (Sciences & Engineering) 79 (2017): 1-23.
19. Ghazaly, Mariam Md, Tawfik Ahmed Yahya, Aliza Che Amran, Zulkeflee Abdullah, Mohd Amran Md Ali, Ahmad Hilkm Jamaludin, and Nursabilillah Mohd Ali. "Force characterization of a tubular linear electromagnetic actuator using finite element analysis method (fem)." Jurnal Teknologi (Sciences & Engineering) 78, no. 11 (2016).
20. Ghazaly, Mariam Md, T. H. Teo, A. Vivek, L. Regeev, A. Kartikesu, L. Vijayan, and M. A. M. Ali. "Point-to-point (PTP) control performances of an upper limb robotic arm." Jurnal Teknologi 78, no. 7 (2016): 131-140.
21. Hussin, Hanipah, Aliza Che Amran, M. A. Hanafiah, Fadzilah Salim, Adlan Ali, Gede Ananta Pramudya, Mohd Razali Yunos, N. M. Yaacob, and M. M. Jaber. "Malaysian teacher/lecturer education development in TVET: A fundamental framework for human capital development." International Business Management 10, no. 15 (2016): 2980-2986.
22. Amran, Aliza Che, Barkan Ugurlu, and Atsuo Kawamura. "Energy and torque efficient zmp-based bipedal walking with varying center of mass height." In 2010 11th IEEE International Workshop on Advanced Motion Control (AMC), pp. 408-413. IEEE, 2010.

K. GOOGLE SCHOLAR-INDEXED PUBLICATIONS – *selected*

h-index: 8, i10-index: 6, Total citations: 158

1. Perumal, N., and Aliza Che Amran. "Automatic load shedding in power system." In Proceedings. National Power Engineering Conference, 2003. PECon 2003., pp. 211-216. IEEE, 2003.
2. Hussin, Hanipah, Aliza Che Amran, M. A. Hanafiah, Fadzilah Salim, Adlan Ali, Gede Ananta Pramudya, Mohd Razali Yunos, N. M. Yaacob, and M. M. Jaber. "Malaysian teacher/lecturer education development in TVET: A fundamental framework for human capital development." International Business Management 10, no. 15 (2016): 2980-2986.
3. Amran, Aliza Che, Gede Pramudya Ananta, Mohd Ariff bin Mat Hanafiah, Adlan Ali, and Che Ku Nuraini Che Ku Mohd. "Development of the Malaysian skills certification for lecturers in tertiary TVET institutions." Journal of Technical Education and Training 12, no. 1 (2020).
4. Amran, Aliza Che, Barkan Ugurlu, and Atsuo Kawamura. "Energy and torque efficient zmp-based bipedal walking with varying center of mass height." In 2010 11th IEEE International Workshop on Advanced Motion Control (AMC), pp. 408-413. IEEE, 2010.
5. Ali, Fariz, Aliza Che Amran, and Atsuo Kawamura. "Bipedal robot walking strategy on inclined surfaces using position and orientation based inverse kinematics algorithm." In 2010 11th International Conference on Control Automation Robotics & Vision, pp. 181-186. IEEE, 2010.
6. Ali, Fariz, Aliza Che Amran, and Atsuo Kawamura. "Slope-walking of a biped robot with position and orientation based inverse kinematics method." In ICCAS 2010, pp. 1724-1728. IEEE, 2010.
7. Saleh, Saleha Mohamad, Aliza Che Amran, Muliati Sedek, Mariam Md Ghazaly, and Muhamad Yusuf Hasan. "Operationalizing Circular Economy Goals through Food-Waste Composting: A GREEN Framework for Hawker Centers."